

CONGESTION CONTROL ALGORITHM

Congestion:

Too many packets present in (a part of) the network causes packet delay and loss that degrades performance. This situation is called congestion.

- The network and transport layers share the responsibility for handling congestion. Since congestion occurs within the network, it is the network layer that directly experiences it and must ultimately determine what to do with the excess packets.
- However, the most effective way to control congestion is to reduce the load that the transport layer is placing on the network. This requires the network and transport layers to work together.

The best way to handle both problems(flow control & Congestion Control) is to get the host to slow down.

Thus, a host can get a “slow down” message either because the receiver cannot handle the load or because the network cannot handle it.

Several techniques can be employed. These include:

1. Warning bit
2. Choke packets
3. Load shedding
4. Random early discard
5. Traffic shaping

The first 3 deal with congestion detection and recovery. The last 2 deal with congestion avoidance

1. **Warning bit:** A warning bit is set in a packet header by a router to indicate congestion in the network. When the receiver gets this packet, it informs the sender to reduce its transmission rate.
2. **Choke packets:** A choke packet is a special control packet sent by a router directly to the sender to signal congestion. It instructs the sender to decrease its data sending rate to avoid further congestion.
3. **Load shedding:** Load shedding is a technique where routers intentionally drop packets when the network is overloaded. This helps reduce congestion and maintain overall network performance.
4. **Random Early Discard (RED):** RED is a congestion avoidance technique where routers randomly drop packets before the queue becomes full. This prevents sudden congestion and helps maintain smoother traffic flow.

Traffic Shaping

1. Another method of congestion control is to “shape” the traffic before it enters the network.
2. Traffic shaping controls the rate at which packets are sent (not just how many). Used in ATM and Integrated Services networks.
3. At connection set-up time, the sender and carrier negotiate a traffic pattern (shape)

Traffic shaping is a mechanism to control the amount and the rate of the traffic sent to the network. Two techniques can shape traffic: leaky bucket and token bucket.

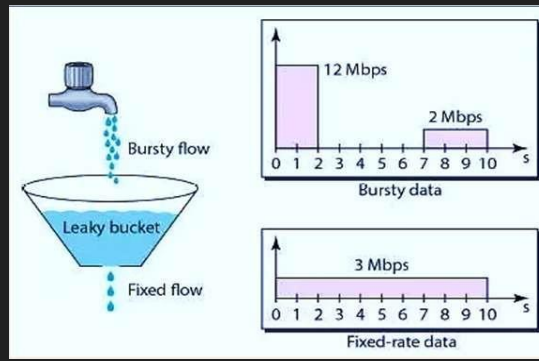
Two traffic shaping algorithms are:

- ✓ Leaky Bucket

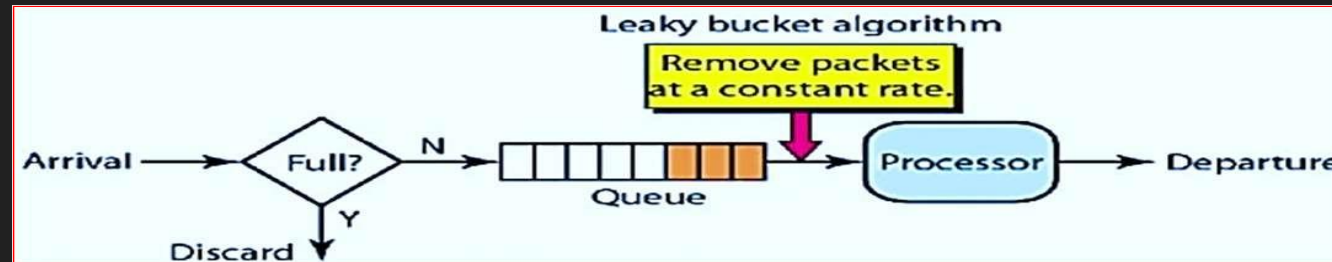
- ✓ Token Bucket

Leaky Bucket

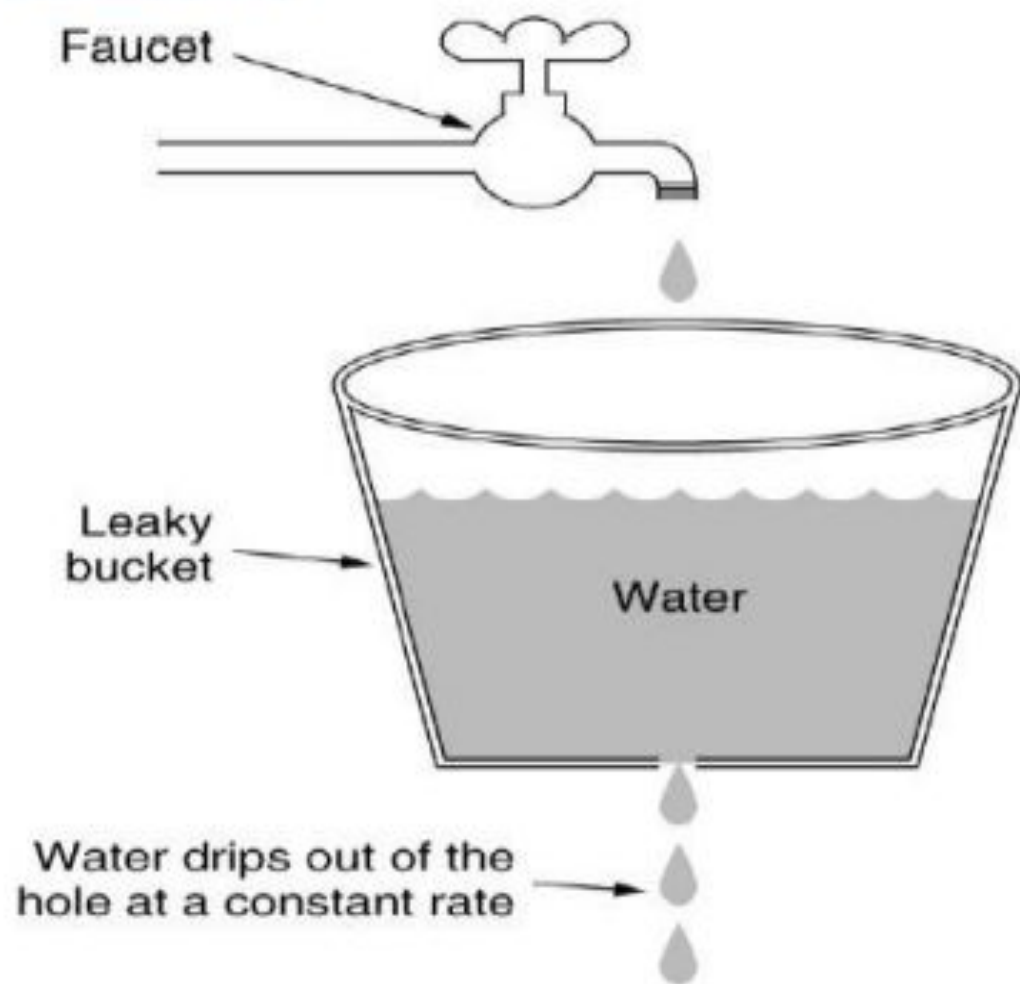
- If a bucket has a small hole at the bottom, the water leaks from the bucket at a constant rate as long as there is water in the bucket.
- The rate at which the water leaks does not depend on the rate at which the water is input to the bucket unless the bucket is empty.
- The input rate can vary, but the output rate remains constant.
- The Leaky Bucket Algorithm used to control rate in a network. It is implemented as a single-server queue with constant service time. If the bucket (buffer) overflows then packets are discarded.



- In Figure , the host sends a burst of data at a rate of 12 Mbps for 2 s, for a total of 24 Mbits of data. The host is silent for 5 s and then sends data at a rate of 2 Mbps for 3 s, for a total of 6 Mbits of data. In all, the host has sent 30 Mbits of data in 10s. The leaky bucket smooth's the traffic by sending out data at a rate of 3 Mbps during the same 10 s.

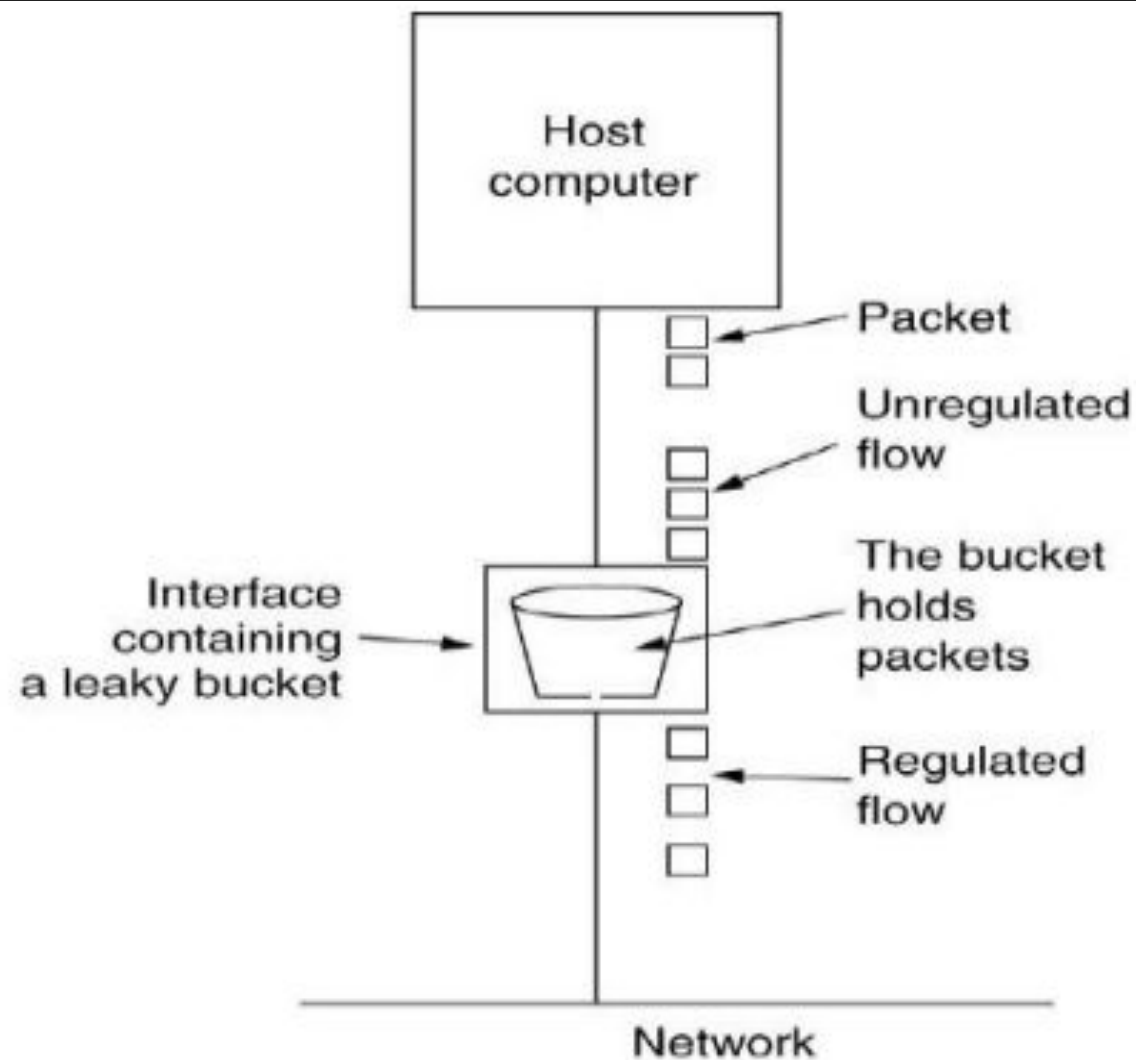


- A FIFO queue holds the packets. If the traffic consists of fixed-size packets (e.g., cells in ATM networks), the process removes a fixed number of packets from the queue at each tick of the clock.
- If the traffic consists of variable-length packets, the fixed output rate must be based on the number of bytes or bits.



(a)

(a) A leaky bucket with water.



(b)

(b) a leaky bucket with packets.

1. The leaky bucket enforces a constant output rate (average rate) regardless of the burstiness of the input. Does nothing when input is idle.

2. The host injects one packet per clock tick onto the network. This results in a uniform flow of packets, smoothing out bursts and reducing congestion.

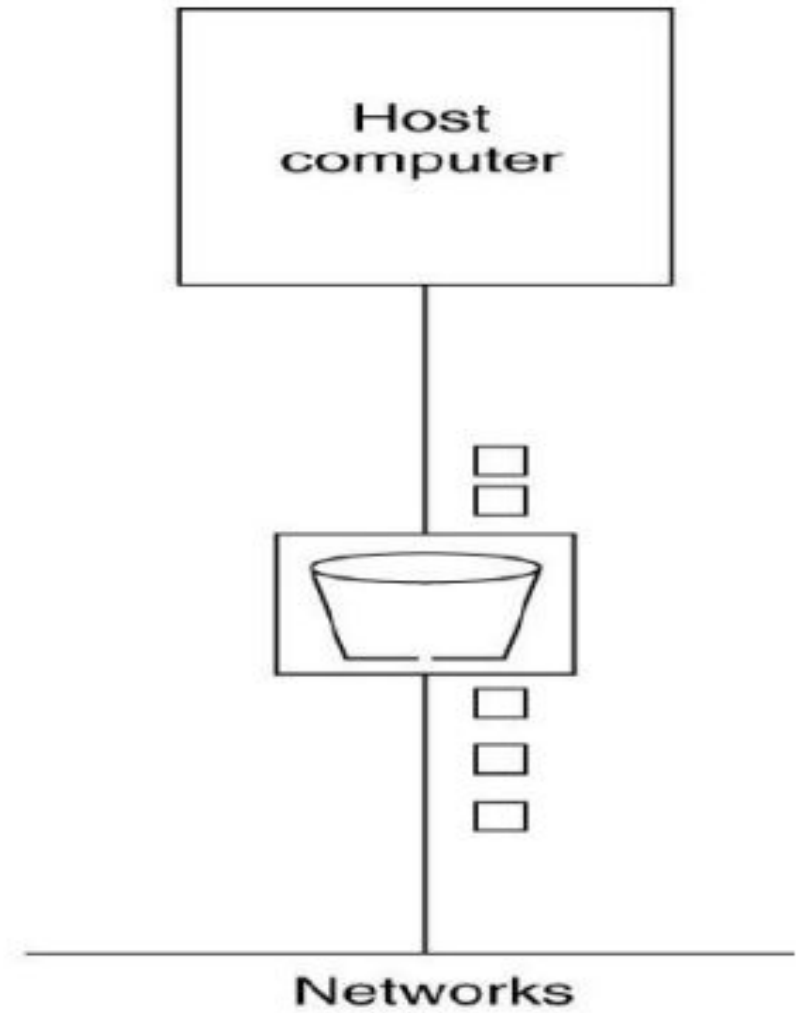
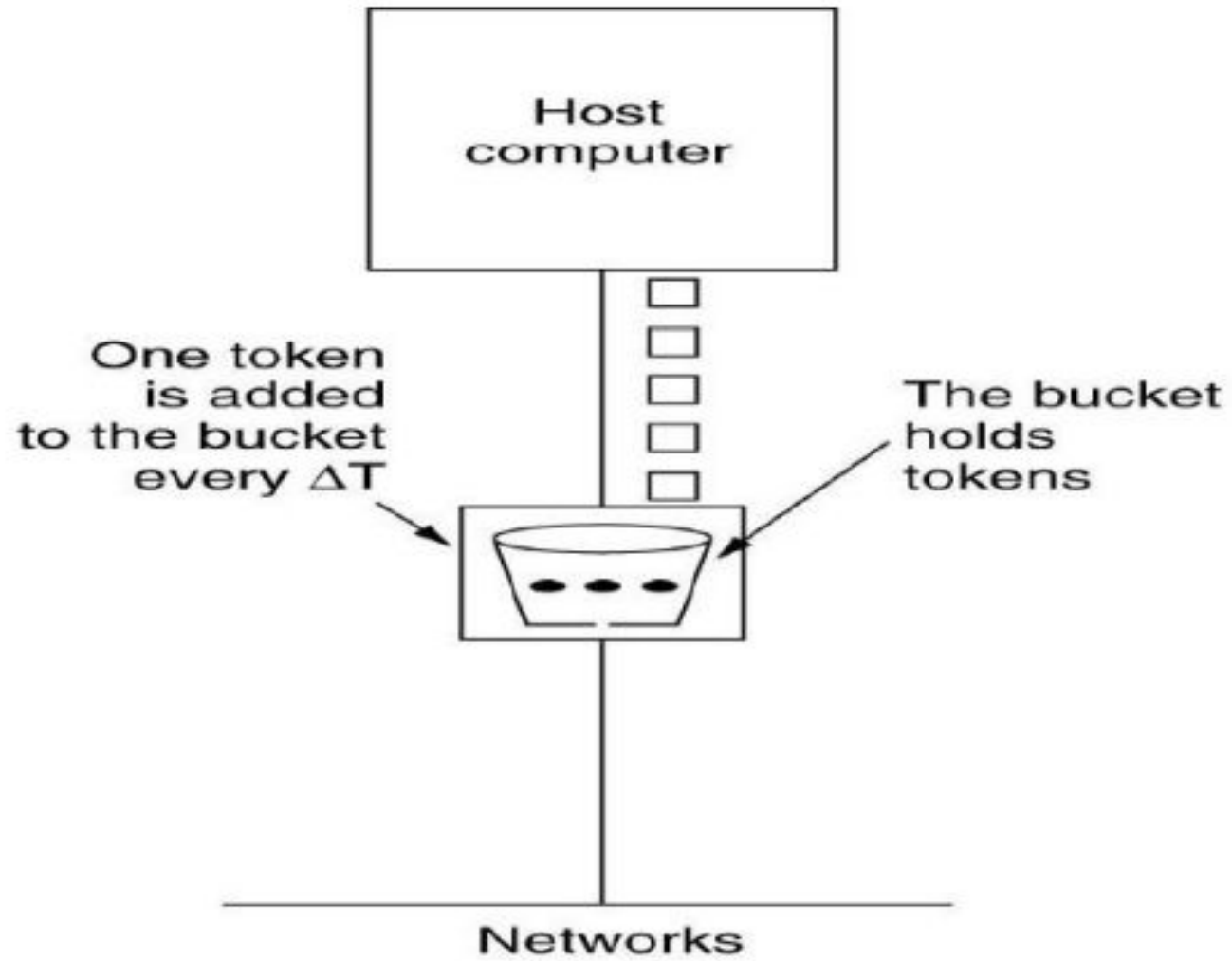
3. When packets are the same size (as in ATM cells), the one packet per tick is okay. For variable length packets though, it is better to allow a fixed number of bytes per tick. E.g. 1024 bytes per tick will allow one 1024-byte packet or two 512-byte packets or four 256-byte packets on 1 tick

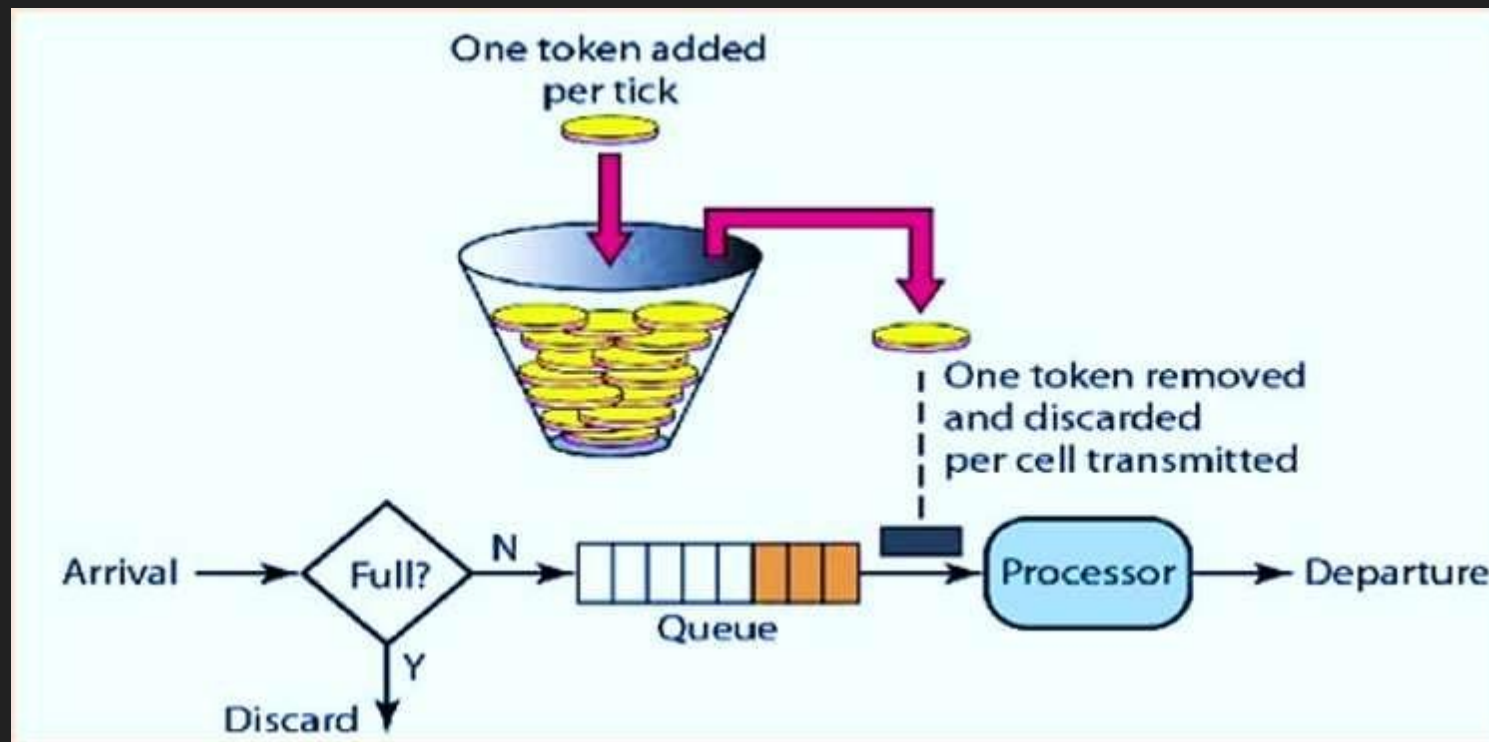
Token Bucket Algorithm

1. In contrast to the LB, the Token Bucket Algorithm, allows the output rate to vary, depending on the size of the burst.
2. In the TB algorithm, the bucket holds tokens. To transmit a packet, the host must capture and destroy one token.
3. Tokens are generated by a clock at the rate of one token every sec.
4. Idle hosts can capture and save up tokens (up to the max. size of the bucket) in order to send larger bursts later.

Token Bucket

- The leaky bucket is very restrictive. It does not credit an idle host. For example, if a host is not sending for a while, its bucket becomes empty.
- Now if the host has bursty data, the leaky bucket allows only an average rate. The time when the host was idle is not taken into account.
- On the other hand, the token bucket algorithm allows idle hosts to accumulate credit for the future in the form of tokens. For each tick of the clock, the system sends n tokens to the bucket.
- The system removes one token for every cell (or byte) of data sent. For example, if n is 100 and the host is idle for 100 ticks, the bucket collects 10,000 tokens.





- The token bucket can easily be implemented with a counter. The token is initialized to zero. Each time a token is added, the counter is incremented by 1. Each time a unit of data is sent, the counter is decremented by 1. When the counter is zero, the host cannot send data.
- The token bucket allows bursty traffic at a regulated maximum rate.

Leaky Bucket vs. Token Bucket

1. LB discards packets; TB does not. TB discards tokens.
2. With TB, a packet can only be transmitted if there are enough tokens to cover its length in bytes.
3. LB sends packets at an average rate. TB allows for large bursts to be sent faster by speeding up the output.
4. TB allows saving up tokens (permissions) to send large bursts. LB does not allow saving.

Leaky Bucket VS Token Bucket

Parameter	Leaky Bucket	Token Bucket
Tokens	Token Independent	Token Dependent
Overflow	If Bucket overflows Packets are discarded	If Bucket overflows Tokens are discarded not Packets
DATA Flow rate	Constant	Variable
Support for Bursty traffic	Not Supported	Supported if sufficient tokens are available
When Application is idle	Does Nothing	Generates the tokens to support bursty traffic
Packet transfer	Packets transfer continuously with constant rate	Packets can be transferred if and only if tokens are available

THANK YOU